S^{3}TRM: Spectral-Spatial Unmixing of Hyperspectral Imagery Based on Sparse Topic Relaxation-Clustering Model

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***Abstract*— Hyperspectral unmixing (HU) has been one of the hot spots in hyperspectral remote sensing research and has great potential in many applications. In recent years, the employment of the probabilistic topic model to mine latent topics in hyperspectral images has been an effective way for spectral unmixing. However, these methods fail to fully exploit the potential of topic models in uncovering image semantics and need extra sparsity constraints, which greatly increases the complexity of the model. In addition, the spatial information which can provide the correlation of features in adjacent pixels is usually ignored in topic model-based HU. To solve these problems, a spectral-spatial unmixing framework of hyperspectral imagery based on a sparse topic relaxation-clustering model (S^{3}TRM) is proposed. In S^{3}TRM, the topic model combined with implied sparse prior constraints are introduced, and the sparse characteristics of S^{3}TRM are used to capture the semantic representation of the spectrum. With the proposed relaxation-clustering strategy, multiple possible spectral representations of features can be obtained, which further alleviates the influence caused by endmember variability. Group clustering is used to locate the endmember quickly and accurately. Moreover, superpixel segmentation is considered to supplement the spatial distribution information of features, thereby improving the fractional abundance. Experiments on a synthetic dataset and three well-known real hyperspectral datasets confirm the excellent performance of the proposed framework in both qualitative assessment and quantitative evaluation, compared with the other state-of-the-art methods.**

***Index Terms*— Hyperspectral unmixing (HU), relaxation-clustering strategy, sparse prior, superpixel segmentation, topic model.**

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I. Introduction

With the development of remote sensing sensors, the wealth of spectral information contained in hyperspectral images (HSIs) has opened new perspectives in many fields, such as environmental monitoring, precision agriculture, forestry monitoring, and mineral exploration [1]. However, due to the mixing effects of the ground surface and the low spatial resolution of the sensor, a single pixel may contain multiple spectral features, which causes the "mixed pixel" problem [2], [3]. In this case, hyperspectral unmixing (HU) is one of the crucial steps for the deep development of HSIs [4]. Specifically, HU is aimed at identifying the spectral signatures of the pure constituent spectra in HSIs (called endmembers) and the corresponding proportion of each endmember (known as an abundance) in each image pixel [5].

Spectral unmixing methods can be grouped into the following categories [6]: 1) geometrical; 2) sparse regression; 3) deep learning; and 4) statistical methods. Geometrical approaches assume that linearly mixed vectors are located in a positive cone or in a simplex set. In this way, the vertex component analysis (VCA) uses convex geometry rules to efficiently perform unmixing tasks [7]. The determination of the fractional abundances can then be handled by least-squares-based methods such as fully constrained least squares (FCLSs), due to a simple structural model, fast computation when the number of endmembers is small [8], [9]. However, the geometrical approaches are mainly suitable for the situation where the image contains pure pixels and meets the noise-free condition. Despite its high computational efficiency, the geometrical models often fail to capture highly mixed spectral features [10]. Facing the highly mixed scenes, the sparse regression method is able to treat the unmixing process as a linear regression problem on a given spectral library. The input image is expressed as a combination of spectral signatures by the semisupervised procedure. For these approaches, the spectral library for image features needs to be constructed, and the regression process needs a higher computational cost [11], [12]. Neural network-based HU methods have been proposed to estimate the abundances of mixed pixels [13]. Recently, nonnegative sparse autoencoder and denoising autoencoder have achieved fast development in obtaining the endmember signatures and abundance fractions

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simultaneously [14]. Although good results can be obtained, they usually are complex to calculate and multiple parameters need to be adjusted [15]–[17].

Holding the advantage of not requiring vast training samples and the prior spectral library, statistical approaches have been widely employed as an unsupervised HU approach. Statistical approaches determine endmembers and abundance parameters by using parameter estimation techniques, thus expressing them in a probability distribution. The core of this approach is to assume that the endmembers of each pixel are generated by sampling a certain probability distribution [18], [19]. Considering the problem of the endmember variability, statistical methods are usually taken as solutions to model the distribution of endmembers. The current popular methods based on statistical model includes nonnegative matrix factorization (NMF) and Gaussian mixture model (GMM) [20], [21]. Due to the property to effectively deal with multiple tasks by expressing a high-level semantic understanding of data, the probabilistic topic model (PTM) has shown great potential in the task of linear HU.

In order to avoid over-fitting caused by complex PTM, sparsity can be used to reduce the solution space and the number of parameters that need to be inferred [22]. It is consistent with the fact that most pixels are mixed only by a few features in the scene for HSI. In detail, Wang and Qi [22] first tackled the unmixing task as a latent topic-based method and proposed the sparsity-constrained probabilistic latent semantic analysis (SC-PLSA), and this algorithm obtained relatively high efficiency in HU. However, this model which introduces sparse constraints by selecting regularization terms with auxiliary parameters may cause problems with large-scale datasets. Fernandez-Beltran *et al.* [23] developed a new PLSA-based model, named dual-depth sparse probabilistic latent semantic analysis (DEpLSA). In DEpLSA, two sparsity factors are introduced into the topic and image to combine sparsity. However, the introduced sparsity factors are only fixed constants artificially preset, and the model shows high model complexity and time consumption.

Among the PTMs, the fully sparse topic model (FSTM) was first proposed in the text field and has been successfully applied to scene classification [24]. It has natural sparse properties and can mine the sparse topics of documents with linear convergence, which is a simple but efficient method for highly complex scenes. However, strong spatial and spectral correlation usually appears in the abundance of pixels. When FSTM is directly applied to HU, it may produce ill-posed conditions of the model and ignore the spatial correlation of adjacent features in HSIs. For HSI processing, studies related to spatial distribution and joint spectral-spatial utilization have been widely applied. For example, considering spectral-spatial information in the classification of hyperspectral and light detection and ranging (LiDAR) data can help improve application performance [25]–[27]. And introducing spatial information to abundance estimation is an effective way to promote the unmixing results [28], [29]. Moreover, due to the complex diversity of ground features and the influence of imaging conditions, the phenomenon of "same objects with different spectra," that is, endmember variability, is widespread in HSIs, resulting in limited accuracy of unmixing images with a fixed set of endmembers.

Motivated by the mentioned problems, in this work, a novel and effective spectral-spatial unmixing framework of hyperspectral imagery based on a sparse topic relaxation-clustering model (S^3TRM) is proposed. In S^3TRM, the natural sparse characteristics of the model are applied to HU to avoid over-fitting. The flexible implicit semantic mining and the sparse reasoning ability of the Frank–Wolf algorithm are used to obtain the endmembers and abundances. Considering that the distribution of features usually exhibits strong spatial and spectral correlation in the adjacent pixels of HSIs, simple linear iterative clustering (SLIC) superpixel segmentation is exploited to obtain homogeneous regions. A weighting strategy for the model to supplement spatial and spectral information is proposed to obtain more accurate abundance. For endmember variability issues, the sparse topic relaxation-clustering model (STRCM) is proposed. The relaxation-clustering strategy of the model is used to approximate multiple possible representations of endmembers [30]. Based on the relaxed topics, fuzzy C-means (FCM) clustering is utilized to acquire the final endmembers [31], [32].

The main contributions of this study are as follows.

1) We proposed the S^3TRM framework to unmix the HSIs based on the mining of natural sparse topics with linear convergence. S^3TRM gives consideration to both time efficiency and satisfactory performance. In S^3TRM, the sparse prior is incorporated into the topic model to reduce the solution space and the number of parameters to be extracted for HU. Integrating the discriminative spatial correlation with the sparse semantics extracted from the initial spectral space, the proposed S^3TRM can effectively solve the incompatibility issues of directly applying topic models to unmixing.
2) Considering the spatial correlation and spectral similarity of features in homogeneous regions, superpixel segmentation strategy is applied to obtain the spatial distribution information of features. Based on the pixel and superpixel matrix, S^3TRM utilizes both the spectral-spatial and semantic information in the complex hyperspectral imagery. The unmixing performance can be effectively improved through the fusion of spatial and spectral information.
3) Based on the proposed relaxation-clustering strategy, STRCM is established to effectively alleviate the influence of endmembers' variability in HSIs. In detail, the multiple representations of each feature can be approximated by the relaxation of endmembers, and the group clustering is exploited to obtain the most representative endmembers.

Comprehensive evaluations on a synthetic dataset and three well-known real HSIs and comparisons with the state-of-the-art HU methods verifies the effectiveness and superiority of the proposed methods.

The rest of the article is arranged as follows. Section II introduces the related work. Section III discusses details about S^3TRM for HSIs unmixing. Section IV is devoted to

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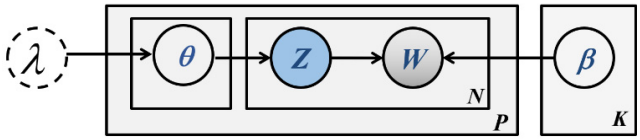


Fig. 1. Probabilistic graphical model of FSTM.

experiments and analysis with hyperspectral datasets. Finally, the conclusions are presented in Section VI.

II. Background

The S^{3}TRM framework is built based on the pixel-level and superpixel-level HU utilizing the topic model. In this section, we briefly introduce the related methods for HU: classic topic model and SLIC superpixel segmentation.

A. Classic Topic Model

The classic topic model includes PLSA and Latent Dirichlet Allocation (LDA) [33]–[35]. In 2001, Hofmann proposed PLSA model. For PLSA, the terms W, Z, and D denote the visual word, the topic, and document, respectively. However, since PLSA lacks a probability function to describe the document, the number of model parameters increases linearly with the size of the documents.

In 2003, the LDA model was proposed, which adds Dirichlet distribution to the topic mixture θ based on the PLSA model. LDA provides a probability function for the discrete implicit topics in PLSA and is regarded as a complete PTM. However, the latent semantics discovered from documents by LDA are often dense, which causes a large number of redundant information and needs more storage space. Researchers tried to reduce the complexity of the model by encoding a spike-and-slap distribution or utilizing regularization to induce sparse topics [36]–[38]. However, they still have some limitations: when dealing with large-scale datasets, auxiliary parameters related to regularization terms require model selection, and they cannot directly weigh the sparsity of solutions with time and quality [24].

In 2012, as a simplified variant of PLSA and LDA, FSTM was proposed. FSTM handles the documents with an implicit sparse prior instead of Dirichlet prior and extra sparsity constraints. In particular, it cancels the endowment of the Dirichlet distribution in LDA. In the case of deleting the observation variable associated with each document, it can also be regarded as a variant of PLSA. Fig. 1 represents the probabilistic graphical model of the FSTM. Existing researches show that compared with other models, FSTM has obvious advantages in terms of learning and inference time, model complexity, and sparsity of implicit representations of documents. In more detail, the inference algorithm converges to the optimal solutions at a linear rate, and the learning algorithm has low complexity, which means that there exists an approximately dimensionless method for dimensionality reduction (DR). Moreover, it provides an effective approach to directly weigh the sparsity of solutions with quality and time [24].

B. SLIC Superpixel Segmentation

Various superpixel segmentation algorithms have been proved to be effective in image processing for HU [29]. SLIC is easy to use, and its only parameter is the number of superpixels required. In our research, SLIC segmentation is employed to divide continuous pixels with similar spectral characteristics in HSIs into homogeneous regions, that is, superpixels. Concretely, starting from the initial grid on the image, by iteratively growing local clusters to create segmentation, the HSI with pixels is converted into a 5-D vector in the CIELAB color space (CIE represents Commission International Eclairage, L represents the luminance, and AB represents the chroma). Then the initial cluster centers are sampled based on a regular grid with pixel intervals. These centers will be moved to nearby locations with the lowest gradients, avoiding centering the superpixels on the edges. Each pixel in the SLIC is only associated with the nearest cluster center, and the search area of the cluster center overlaps with its position, and the search area of this area is set near the center of the superpixel. Since the hexagonal grid can reduce the distance distortion of the boundary pixels, this study uses the hexagonal grid as the initial cluster to obtain a more uniform homogeneous area.

III. Spectral-Spatial Unmixing Framework Based on STRCM

To fully consider the applicability of the unmixing model and effectively utilize the semantic information in image, the S^{3}TRM framework is proposed for HSIs unmixing. This is an ensemble approach combining spectral, spatial, and semantic information. For S^{3}TRM, four tasks have to be addressed: 1) spectral and spatial matrix generation; 2) object-level topic unmixing; 3) pixel-level topic unmixing; and 4) spectral-spatial semantic unmixing. The overall flowchart of HU based on the S^{3}TRM framework is presented in Fig. 2. In the flowchart, the unmixing task is performed at the pixel-level and object-level, respectively. The final abundances are determined by the adaptive combination of spatial abundances and spectral abundances. Thus, more accurate unmixing results incorporating spatial-spectral information are obtained.

A. Spectral and Spatial Matrix Generation

Rich spectral information of HSIs provides data support for feature classification and target recognition. In the homogeneous regions of HSIs, the abundance vectors of pixels usually show strong spatial and spectral correlation. In order to make full use of the spatial distribution information of the features, superpixel segmentation is used to improve the abundance results. Before integrating the vectors, the uniform method is utilized to split the whole HSI into pixel or superpixel blocks. Pixels and superpixels are used as basic units to generate corresponding object-level and pixel-level topic matrices, respectively. The indexes of the individual bands of each pixel or superpixel correspond to different visual words, and the values of different reflectance are similar to the number of individual words in the topic model. Based on this, the hyperspectral cube is converted into a large number of

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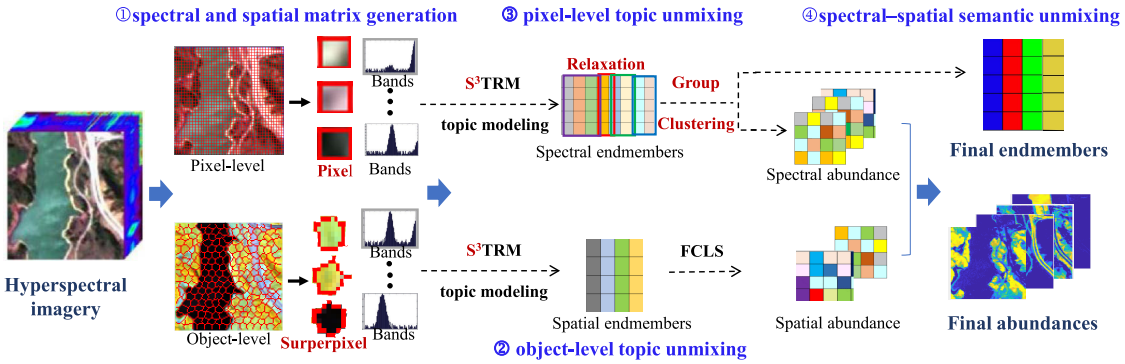


Fig. 2. Flowchart of HU based on the S³TRM framework. Pixel-reflectance matrix and superpixel-reflectance matrix are first constructed at pixel-level and object-level, respectively. For object-level HU, the superpixel-reflectance matrix is input into the STRCM to obtain spatial endmembers, then FCLS is applied to determine the abundances with spatial information. For pixel-level HU, the pixel-reflectance matrix is modeled through the relaxation strategy of STRCM to estimate the spectral endmembers, and the group clustering strategy is adopted to obtain abundance results with spectral information. The final abundances are determined by the adaptive combination of spatial abundances and spectral abundances.

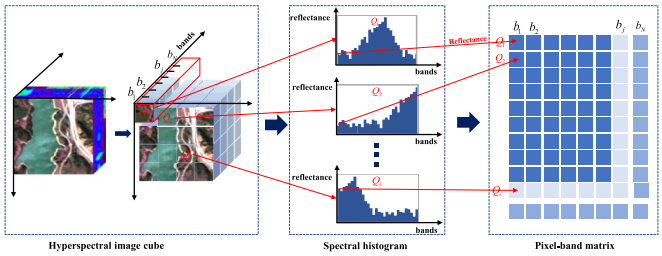


Fig. 3. Generation of the pixel-band matrix.

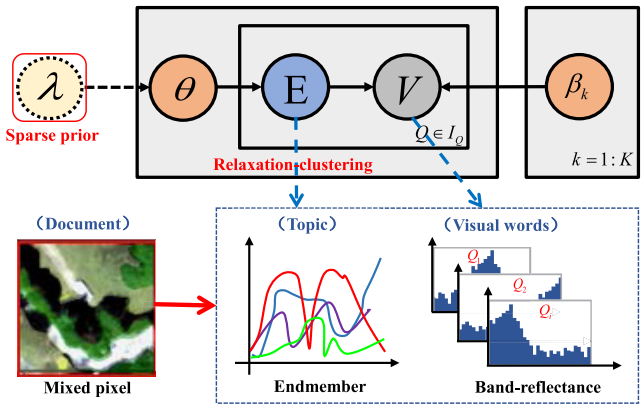


Fig. 4. Correspondence between HU and STRCM.

pixel-based or superpixel-based spectral histograms, and then the pixel-band matrix of the HSI is generated.

As shown in Fig. 3, each row of the matrix represents a pixel or superpixel, and each element denotes the value of the spectral reflectance in each band. When STRCM is applied to the HSIs unmixing, the relationship between their respective terminologies is described in Fig. 4. A corpus corresponds to an HSI, and a document is equivalent to one superpixel/pixel Q in the image. Topics depict the endmembers E. The word corresponds to the spectral band b_j, and the frequency of the term in every document is equivalent to the band reflectance V_j. Specifically,

b_N: Set of band indices of superpixel/pixel Q, i.e., each element in b_N is the index of a band appearing in Q.

j: The band index of superpixel/pixel Q, expressed as $\{1, 2, \ldots, N\}$.

V_j: The band reflectance of superpixel/pixel Q, which is equivalent to the frequency of the term in every document. V_{ij} denotes the reflectance of the ith superpixel/pixel at the jth bands, and V_j^k indicates the reflectance of endmember E_k at the jth bands.

Q: A superpixel/pixel represented as a band reflectance vector, $Q_i = (V_{ij})_{j \in b_N}$, where i is the index of superpixels/pixels.

I_Q: The image consisting of τ superpixels/pixels, $I_Q = \{Q_1, \ldots, Q_\tau\}$.

K: Number of endmembers.

B. Object-Level Topic Unmixing Based on S³TRM Framework

S³TRM framework requires few arithmetic operations, linearly related to the number of bands or/and the number of endmembers to be inferred. It approximates the probability distribution corresponding to the original data by generating two distribution functions, and provides a new approach for spectral unmixing based on linear spectral mixing models.

At the object level, the superpixel reflectance matrix generated in the previous step is input to STRCM. The index of the N bands is represented as j, the index of superpixel Q is expressed as i, and the index of the topics is represented by k. STRCM assumes that a mixed superpixel can be represented as K endmembers, denoted by $\beta_1, \ldots, \beta_K$. The unmixing process of STRCM for the HSIs is shown as follows:

1) For each superpixel, randomly select a K-dimensional topic proportion θ within a reasonable range.
2) For each band in superpixel Q, a latent endmember E_k is selected for each superpixel conditionally to the probability $p(E_k|Q) = \theta_k$, then generate spectral band b_j with probability $p(b_j|E_k) = \beta_{kj}$. The probability $p(b_j|Q)$ between the spectral band b_j and superpixel

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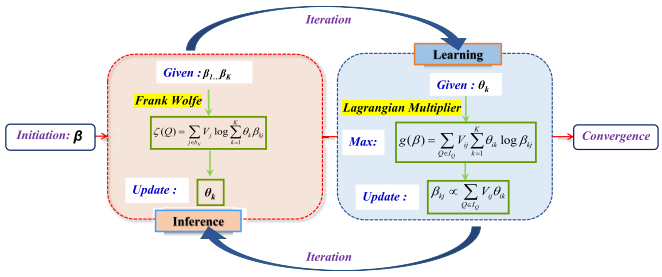


Fig. 5. Inference and learning task of STRCM.

Q can be described as follows:

$$P(b_j|Q) = \sum_{k=1}^{K} p(b_j|E_k)p(E_k|Q). \tag{1}$$

It is worth noting that despite being a simplified variant, an implicit prior about θ that subject to (2) has been confirmed to exist in STRCM

$$P(\theta|\lambda) \propto \exp(-\lambda \cdot \| \theta \|_0). \tag{2}$$

This interesting property of "implicit modeling" enables STRCM to converge to the best solution at a linear rate. In the case of solutions with at most t nonzero components, it models one's solution to the problem $\max_{\theta \in \Delta_1}\{f(\theta) : ||\theta||_0 \leq t\}$, where Δ_1 represents the unit simplex in R^K, and $||\theta||_0$ denotes the number of nonzero terms of θ. In fact, for some constant λ, i.e., the constraint $||\theta||_0 \leq t$ can be seen as adding a penalty term $\lambda \cdot ||\theta||_0$ to the objective function. In this research, STRCM is used for modeling the endmembers with sparsity.

To obtain sparse implicit endmembers, STRCM will use the sparsity of original pixels and endmember proportions. In other words, learning endmembers is equivalent to the multiplication of the new representation of the image and the matrix representation of training pixels. The inference and learning tasks of the STRCM algorithm are shown in Fig. 5.

The inference task is reconstructed as a concave maximization problem on the simplex of endmembers. Moreover, the probability that spectral band b_j appears in a given superpixel Q can be denoted by the following equation:

$$P(b_j|Q) = \sum_{k=1}^{K} P(b_j|E_k) \cdot P(E_k|Q) = \sum_{k=1}^{K} \theta_k \beta_{kj}. \tag{3}$$

Broadly speaking, as in (4), the inference task can be described as the problem of seeking θ to maximize

$$\begin{aligned} \zeta(Q) &= \log \prod_{j \in b_N} P(b_j|Q)^{V_j} \\ &= \sum_{j \in b_N} V_j \log P(b_j|Q) \\ &= \sum_{j \in b_N} V_j \log \sum_{k=1}^{K} \theta_k \beta_{kj}. \end{aligned} \tag{4}$$

Especially, denoting as $x_j = \sum_{k=1}^{K} \theta_k \beta_{kj}$ and $x = (x_1, \ldots, x_V)^t$. The objective function (4) can be expressed as follows:

$$\zeta(Q) = \sum_{j \in b_N} V_j \log x_j. \tag{5}$$

Thus, the optimization of θ is transformed into the optimization of x. Note the following equation:

$$x = (x_1, \ldots, x_V)^t = \sum_{k=1}^{K} \theta_k \beta_k. \tag{6}$$

Considering the fact that $\sum_k \theta_k = 1$, $\theta_k \geq 0$ we can know that x represents the convex combination of K endmembers $\beta_1, \ldots, \beta_K$. For simplex $\Delta = \text{conv}(\beta_1, \ldots, \beta_K)$, which implies $x \in \Delta$. Therefore, the inference task is again the problem of searching for $x \in \Delta$ to maximize the objective function (5). Moreover, only limiting the number of iterations in the Frank–Wolfe algorithm can directly control the sparsity of the endmember proportions θ. For the stopping rule, the maximum iterations and the convergence criterion are adopted in STRCM. Specifically, the predefined maximum iteration number is 100. In addition, the convergence criterion for inference is set to 1e-6.

The learning task of STRCM is to learn all endmembers β, given an image I_Q. Electromagnetic (EM) algorithm is used iteratively to learn the model. In this case, the E-step and M-step steps will be repeated until convergence. The topic proportion denoted by θ_k for superpixel $Q \in I_Q$ has been inferred in step E. In detail, for E-step: do inference for every superpixel Q of image I_Q; For M-step: we have to maximize objective function (7) with regard to β

$$\begin{aligned} \zeta(I_Q) &= \sum_{Q \in I_Q} \zeta(Q) \\ &= \sum_{Q \in I_Q} \sum_{j \in b_N} V_{ij} \log \sum_{k=1}^{K} \theta_{ik} \beta_{kj} \\ &\geq \sum_{Q \in I_Q} \sum_{j \in b_N} V_{ij} \sum_{k=1}^{K} \theta_{ik} \log \beta_{kj}. \end{aligned} \tag{7}$$

Combining this fact with the following equation:

$$\sum_{k=1}^{K} \theta_{ik} = 1, \quad \theta_{ik} \geq 0 \ \forall k. \tag{8}$$

The last term was derived using Jensen's inequality. Then we need to maximize the following equation:

$$g(\beta) = \sum_{Q \in I_Q} \sum_{j \in b_N} V_{ij} \sum_{k=1}^{K} \theta_{ik} \log \beta_{kj}. \tag{9}$$

Considering the Lagrange function and forcing its derivatives to be 0, this learning can update all endmembers only by performing simple calculations according to (10). Based on this, the endmembers with spatial information are acquired

$$\beta_{kj} \propto \sum_{Q \in I_Q} V_{ij} \theta_{ik}. \tag{10}$$

The FCLSs algorithm is used to obtain the abundance after obtaining the endmembers. Abundance sum-to-one constraint (ASC) and abundance nonnegative constraint (ANC)

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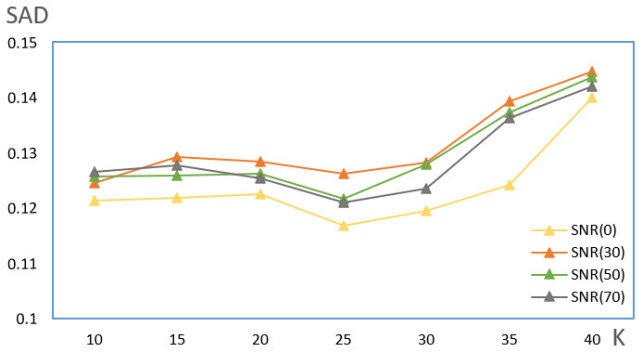


Fig. 6. Sensitivity analysis on topic number K for Simu dataset.

constraints are considered simultaneously in this algorithm, which obtains the abundance matrix by the idea of calculating the minimum value of the sum of squares of errors.

C. Pixel-Level Topic Unmixing Based on S^{3}TRM Framework

Similar to the acquisition process of spatial endmembers, at the pixel level, the pixel reflectance matrix is input to the STRCM to determine spectral endmembers. For the real scenes, one spectrum does not completely correspond to one material in HSIs. This phenomenon means that the endmembers can change with the pixel. To avoid the effect of unmixing caused by endmembers' variability, the sparsity of the STRCM is used to relax the topics to obtain the possible forms of each endmember. In other words, the topics are more than the actual number of endmembers when using STRCM for unmixing. To explore the effect of the relaxation of the latent topic on inaccurate endmember estimation, the Simu dataset with nine endmembers is used for testing. In our experiments, the number of topics K is set to 10, 15, 20, 25, 30 for Simu datasets with different signal-to-noise ratios (SNRs). As can be shown in Fig. 6, when K is set below 30, the change of the topic number has little effect on the endmember extraction. The proposed method is relatively stable when the number of topics varies within a reasonable range. Once the number of topics exceeds a reasonable range, the accuracy gradually deteriorates.

In this case, the FCM clustering is introduced to extract endmember. The class membership values, which are output by FCM, may be interpreted as degrees of sharing or posterior probabilities [39]. It allows analysts to adjust the fuzziness of the output and is attractive for decomposing the class composition of pixels. Note also that these values are relative measures of category membership strength, so the calculation is performed for all classifications. Suppose the sample set $S = \{s_1, s_2, \ldots, s_N\}$, s_j represents the D-dimensional feature vector in FCM. The FCM minimizes the objective function denoted as (11) by an iterative optimization, then these samples are clustered into C classes

$$J = \sum_{i=1}^{C} \sum_{j=1}^{N} \mu_{ij}^{m} \parallel s_j - v_i \parallel^2 . \tag{11}$$

The constraints are as follows:

$$\mu_{ij} \in [0, 1];\ 0 < \sum_{j=1}^{N} \mu_{ij} < N\ \forall i;\ \sum_{i=1}^{C} \mu_{ij} = 1\ \forall j \tag{12}$$

where μ_{ij} represents the fuzzy membership of pixel s_j in the ith cluster; m is fuzziness parameter; C denotes the number of classes; v_i represents the ith cluster center; and $m > 1$ denotes the weighted index of each fuzzy membership used to control the fuzziness of the analysis. They obtain the clustering centers and the membership degrees in an unsupervised clustering manner in the fuzzy model, where the clustering centers refer to endmembers. In this case, the endmembers' generative process can be described as follows: first, based on the sparse reasoning ability and fast convergence characteristics of STRCM, initial possible endmembers are mined in HSIs. Then according to the FCM algorithm, the improved endmembers that are expressed as cluster centers are chosen by deriving the strength of membership of every endmember to the corresponding class. Correspondingly, we can group the corresponding abundances of the same feature according to the division of endmembers, and follow the ASC to obtain spectral abundances.

D. Spectral-Spatial Semantic Unmixing

Based on the ability of flexible implicit semantic mining and the Frank–Wolf algorithm's sparse reasoning, endmembers and abundances can be quickly estimated. However, the sparse topics obtained only by STRCM may lack representative semantic features. In addition, existing unmixing studies have shown that different models show special performance according to their characteristics in HU. For real HSI, the endmembers may have multiple representations, whose shape cannot be represented by single distribution. In this case, STRCM with sparse relaxation and group clustering can approximate any distribution found in the library, and has higher accuracy and flexibility in solving the endmember variability. The abundance vector shows a strong spatial relationship in homogeneous regions of spatial adjacency and spectral similarity. By incorporating the spatial distribution characteristics of the features, the superpixel segmentation strategy can obtain more accurate abundance. Based on this, abundances of the same features obtained from STRCM with spectral information and STRCM with spatial information are united by appropriate weights to achieve final abundances.

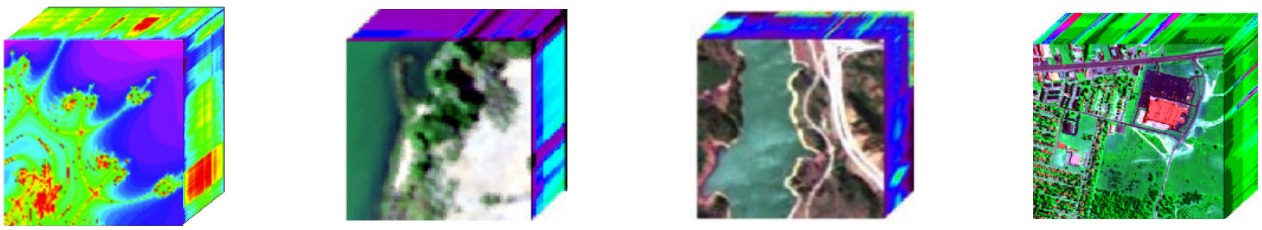
Finally, the pseudocode of the S^{3}TRM algorithm is summarized in Algorithm 1.

IV. Experiments and Analysis

A. Experimental Settings

In this section, a Simu dataset and three real hyperspectral datasets as shown in Fig. 7 have been considered in our experiments, i.e., Simu dataset, Samson dataset, Jasper dataset, and Urban dataset. They are popular and widely used datasets in HU; corresponding datasets and the ground-truth are acquired through the approach provided in [40].

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(a) (b) (c) (d)

Fig. 7. HSIs used in experiments. (a) Simu. (b) Samson. (c) Jasper. (d) Urban.

Algorithm 1 S^{3}TRM for HU

Input: hyperspectral dataset Y.
Output: endmembers E, abundances A.
Step 1./∗ ***Spectral and spatial matrix generation*** ∗/
1.Generate M' and M''.
M' : the pixel-reflectance matrix
M'' : the superpixel-reflectance matrix
Step 2./∗ ***Object-level topic unmixing by*** Fig.4∗/
2. infer the proportion θ_k.
3. learn all topics β by EM scheme.
repeat
a) E-step: Update θ_k by (4) and (5).
b) M-step: Update β_{kj} via (10).
until convergence
4. Obtain spatial abundances A' by FCLS.
Step 3./∗ ***Pixel-level topic unmixing*** ∗/
5. Obtain spectral endmembers E' by STRCM.
6. Generate final endmembers E by (11).
7. Generate spectral abundances A'' via sparse relaxation and group clustering.
Step 4. /∗ ***Spectral-spatial semantic unmixing*** ∗/
8. Generate final abundances A by combining Spectral-spatial information.

In order to evaluate the performance of unmixing, two quantitative metrics are used in our experiments [41]–[44]: the spectral angle distance (SAD) written by (13) and the root-mean-squared error (RMSE), as shown in the following equation:

$$\mathrm{SAD}(\hat{V}, V) = \frac{1}{K}\sum_{i=1}^{K} \arccos \frac{\hat{V}_i \cdot V_i}{\| \hat{V}_i \| \| V_i \|} \tag{13}$$

$$\mathrm{RMSE}(\hat{M}, M) = \sqrt{\frac{1}{R \cdot C}\sum_{i=1}^{R}\sum_{j=1}^{C}(\hat{M}_{i,j} - M_{i,j})^2} \tag{14}$$

where K indicates the number of endmembers. $\hat{V}$, V represents estimated endmembers and the corresponding ground-truth. R and C are the size of input image. $\hat{M}$, M act as the estimated abundances and the corresponding ground-truth. To evaluate the performance of S^{3}TRM, the experimental results obtained with the conventional GMM [20], VCA-FCLS [7], minimum volume simplex analysis (MVSA) [9], and STRCM are provided for comparison. In addition, we also give the test results obtained with NMF-based methods: NMF-sp [45], collaborative NMF (CNMF) [46], and NMF quadratic minimum volume (NMF-QMV) [47]. And topic-based methods: PLSA [34], LDA [33] and DEpLSA [23]. To further illustrate the effectiveness of the proposed method, the deep learning-based methods—deep autoencoder network (DAEN) [17], stacked nonnegative sparse autoencoder (SNSA) [15], and hyperspectral unmixing using deep image prior (UnDIP) [48]—are shown in the results.

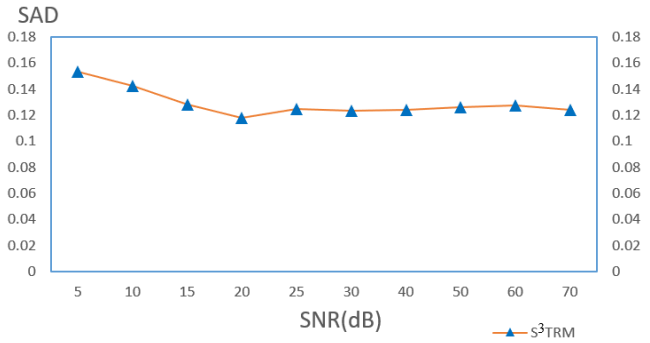


Fig. 8. SAD of the S^{3}TRM with different Gaussian noise.

B. Synthetic Data Experiments With Simu Data

The Simu dataset was generated by the simulation procedure proposed [49], as shown in Fig. 7(a), and has a size of 100 × 100 pixels with 221 channels. It consists of nine endmembers chosen from the geological survey (USGS) spectral library: Kaolinite KGa-1, Dumortierite, Nontronite, Alunite, Sphene, Pyrophyllite, Halloysite, Muscovite, and Kaolinite CM9.

1) Robustness Analysis to Noise: Considering that in a real hyperspectral data production scenario, images are usually affected by many different types of distributions and image corrections where it will also introduce some noise. we conducted this experiment to compare and analyze the performance of algorithms. In this experiment, the Simu dataset was polluted by different levels of white Gaussian noise, where the SNR = {5, 15, 20, 25, 30, 40, 50, 60, 70 dB}, as shown in the following equation:

$$\mathrm{SNR} = 10\log_{10}\frac{E[\mathrm{X}^T\mathrm{X}]}{E[\mathrm{n}^T\mathrm{n}]} \tag{15}$$

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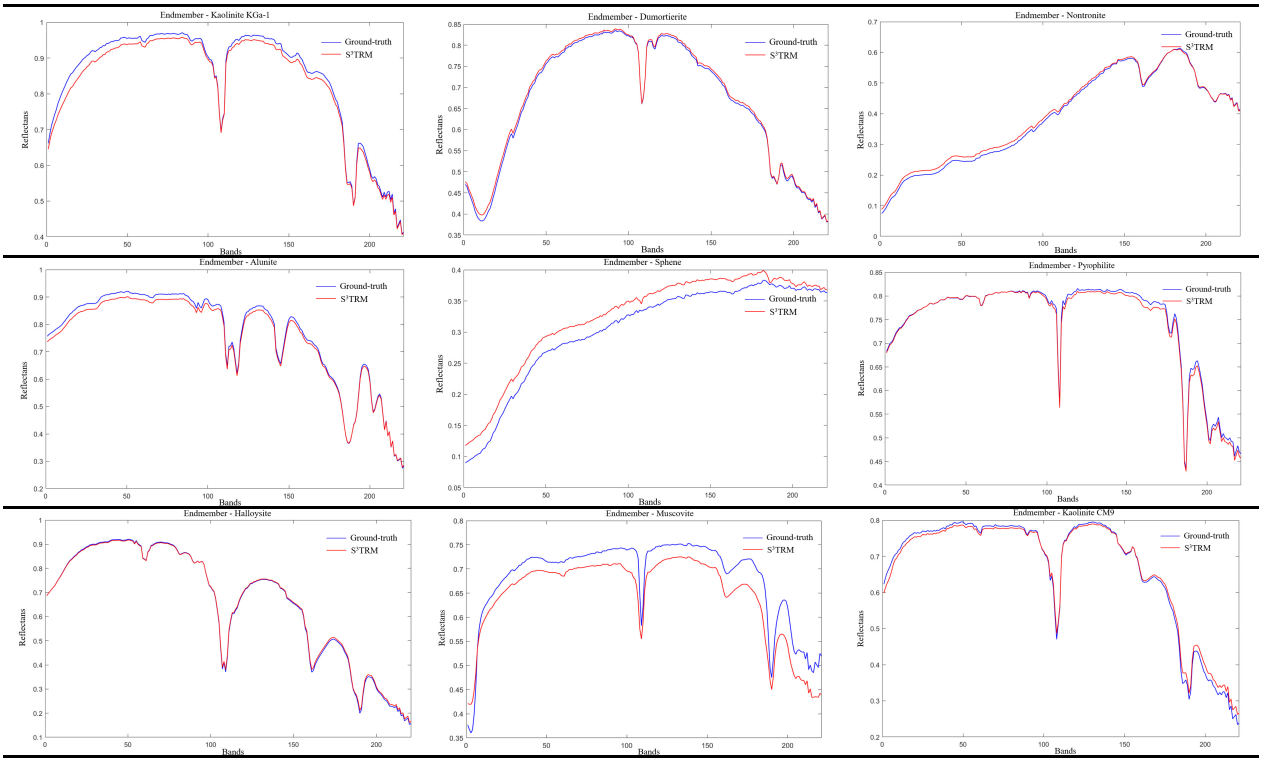


Fig. 9. Endmember results for Simu dataset estimated by S^{3}TRM and ground-truth.

TABLE I

SAD ENDMEMBER ASSESSMENT AND RMSE ABUNDANCE ASSESSMENT FOR THE SAMSON DATASET WITH THREE ENDMEMBERS. NOTE THAT THE EXPERIMENTAL RESULTS SHOW BOTH NONTOPIC-BASED AND TOPIC-BASED METHODS, BEST RESULTS IN BOLD, RESPECTIVELY

Samson	**Non-topic-based methods**								
		VCA-FCLS	MVSA	NMF-sp	NMF-QMV	GMM	DAEN	SNSA	UnDIP
	SAD	0.0713	0.1632	0.1310	0.5135	0.1154	0.2047	0.2029	**0.0648**
	RMSE	0.1946	0.3799	**0.1606**	0.2845	0.2126	0.6097	0.6143	0.3151
	Topic-based methods								
		LDA		PLSA		STRCM		S^{3}TRM	
	SAD	0.3495		0.1927		0.2046		**0.1919**	
	RMSE	0.2773		0.1951		0.2322		**0.1592**	

where $E[\cdot]$ is the expectation operator, X and n denote the original signal and the corresponding noise, respectively. Fig. 8 shows the SAD comparison of the S^{3}TRM with different Gaussian noise.

As can be seen from Fig. 8, as the SNR is increased (e.g., SNR $\leq$ 20 dB), the accuracy of S^{3}TRM displays a slow increase trend. When SNR $\geq$ 20 dB, the curve of the model has less fluctuation under different SNRs. Research has shown that the SNRs observed in real hyperspectral imaging instruments are increasing. Therefore, the consideration of low SNR value should not be taken as an indication of the true parameter value of the current hyperspectral imaging state [49]. Thus, the S^{3}TRM seems to be largely unaffected by the added noise and presents a relatively stable and robust performance under different noise conditions.

Moreover, some endmember visual results are provided as a qualitative evaluation of the proposed HU method for Simu dataset. The endmember results for the proposed S^{3}TRM model are presented in Fig. 9. It can be seen that the S^{3}TRM method achieves almost identical results with the real endmember curves for each feature, further illustrating the effectiveness of the method.

C. Real-Data Experiments With Samson Data

The Samson dataset [50], as shown in Fig. 7(b), with 156 channels and a size of 952 $\times$ 952 pixels, was used for the HU experiment. A size of 95 $\times$ 95 pixels has been selected from the (252, 332)th pixel in the original image. Soil, tree, and water were chosen as the references for three endmembers. The comparison of endmembers and abundances with the different methods is provided in Table I. Fig. 10 shows the obtained abundance by the above methods and ground-truth abundances for the Samson dataset.

Samson is simple and there are numerous pure pixels in the datasets. As can be seen from Table I, VCA-FCLS and UnDIP generate good endmember extraction results for the Samson dataset. VCA-FCLS shows good capabilities to estimate

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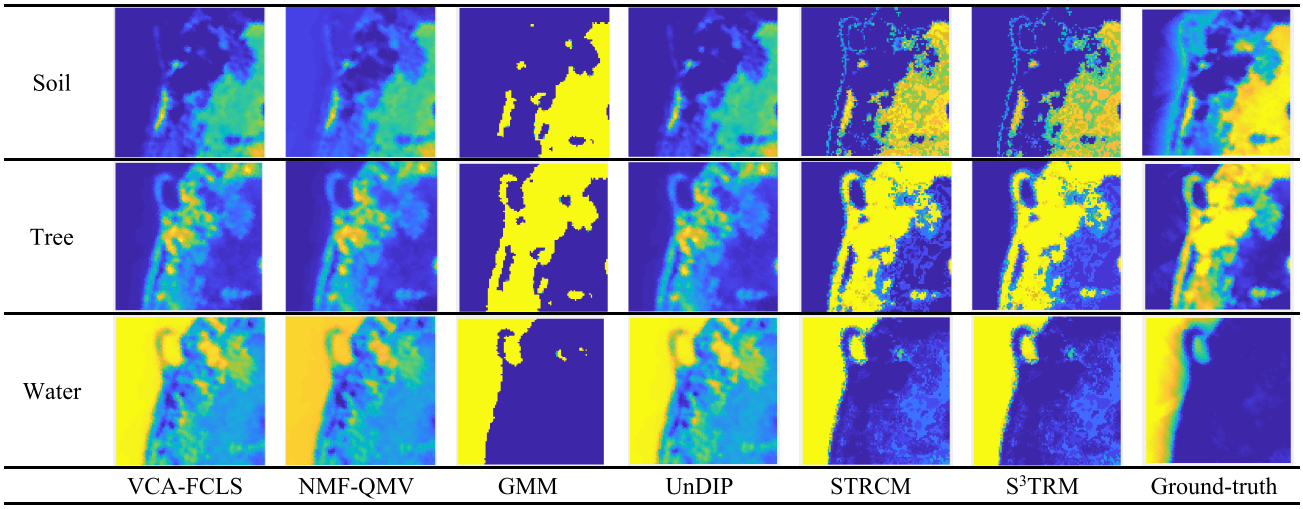


Fig. 10. Abundance results for Samson estimated by different unmixing methods and ground-truth.

TABLE II

SAD ENDMEMBER ASSESSMENT AND RMSE ABUNDANCE ASSESSMENT FOR THE JASPER DATASET WITH FOUR ENDMEMBERS. NOTE THAT THE EXPERIMENTAL RESULTS SHOW BOTH NONTOPIC-BASED AND TOPIC-BASED METHODS, BEST RESULTS IN BOLD, RESPECTIVELY

		Non-topic-based methods							
		VCA-FCLS	MVSA	NMF-sp	NMF-QMV	GMM	DAEN	SNSA	UnDIP
Jasper	SAD	0.1672	0.2842	0.2570	0.4954	0.2315	0.1680	**0.1508**	0.1981
	RMSE	**0.1515**	0.3880	0.1562	0.1647	0.1539	0.1822	0.1798	0.2376
		Topic-based methods							
		LDA		PLSA		STRCM		DEpLSA	S^{3}TRM
	SAD	0.3639		0.3041		0.1943		0.1412	**0.1400**
	RMSE	0.2161		0.2447		0.1489		0.1204	**0.1168**

endmembers for the simple dataset with pure pixels. UnDIP obtains the best SAD but has poor performance in terms of abundance estimation, which may be due to its preextraction of endmembers by simplex volume maximization (SiVM) and holds endmembers fixed to estimate the abundance. Although the endmember results of S^{3}TRM are low compared to the representative VCA, S^{3}TRM has greater robustness for both complex and simple datasets. When considering the RMSE metric, the result provided by S^{3}TRM tends to be more robust as well as accurate than nontopic-based methods. In topic-based methods, the accuracy of S^{3}TRM with sparse relaxation and group clustering strategy achieved the best results in both endmembers and abundances. Moreover, a visual comparison in Fig. 10 shows that the results of S^{3}TRM agree with the ground-truth in the abundance maps and capture more detailed information compared to other unmixing methods.

D. Real-Data Experiments With Jasper Data

Jasper is a popular hyperspectral dataset [50]–[52]. It contains 224 bands and a spectral resolution is 9.46 nm. Moreover, to avoid atmospheric effects, bands 1–3, 108–112, 154–166, and 220–224 have been removed (198 channels remain). As shown in Fig. 7(c), considering the high complexity of the image, the 100 × 100 × 198 subimage cropped from the original image is used in our experiments. The image has the following four endmembers: tree, water, soil, and road. Table II presents the results of the quantitative evaluation of endmembers and abundances. In Fig. 11, the corresponding abundances and the ground-truth abundances are shown.

As can be seen from the results given in Table II, S^{3}TRM achieved competitive results compared with other methods. Specifically, for nontopic-based methods, SNSA and VCA-FCLS achieved the highest performance in terms of endmember and abundance estimation, respectively. S^{3}TRM performs better than other topic-based methods thanks to the consideration of the relaxation-clustering strategy and spectral-spatial information of HSIs. In general, the S^{3}TRM model clearly exhibits the highest unmixing accuracy on both SAD and RMSE. In the case of the Jasper dataset (Fig. 9), tree, water, soil, and road endmembers are represented in different shades of yellow. Compared with other methods, the abundances estimated from S^{3}TRM for each feature match best with regard to the corresponding ones in the ground-truth in visual comparison. Other methods, e.g., VCA-FCLS, NMF-QMV, GMM, and UnDIP, present the problem of apparently massive confusion of features, especially for trees and water.

E. Real-Data Experiments With Urban Data

The Urban dataset [50]–[52], as shown in Fig. 7(d), has a size of 307 × 307 pixels and 210 bands covering the 400–2500 nm wavelength range. In preprocessing, some noisy and corrupted bands were removed (162 bands remain). The ground-truth contains four endmembers, which are asphalt, grass, tree, roof, respectively. Due to the presence of outliers and noise in the Urban dataset, which may cause

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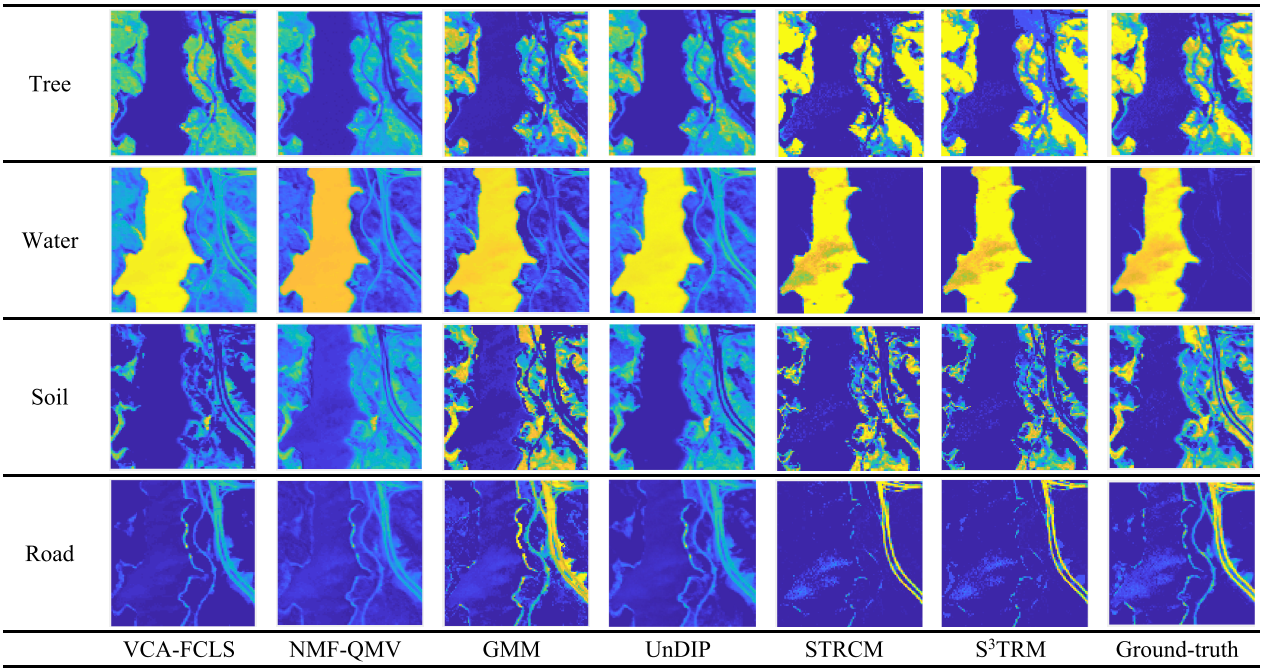


Fig. 11. Abundance results for Jasper estimated by different unmixing methods and ground-truth.

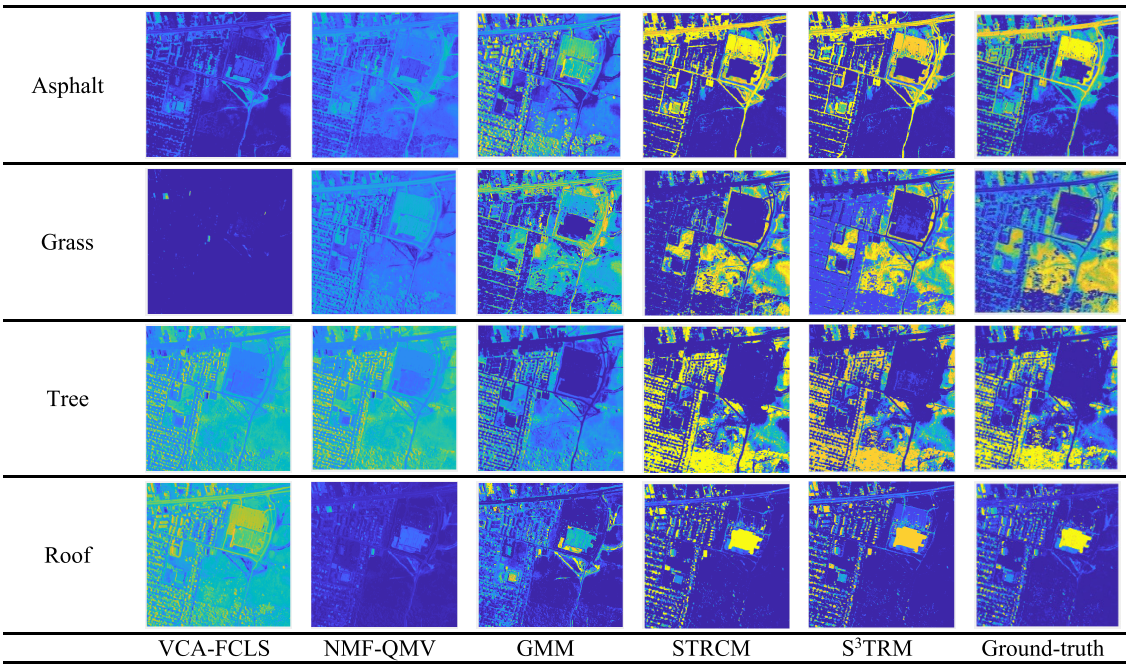


Fig. 12. Abundance results for Urban estimated by different unmixing methods and ground-truth.

difficulties for UnDIP in extracting endmembers, we do not show its results in the experiments. Fig. 12 plots the comparison of the abundance results obtained by the ground-truth with the previous methods including VCA-FCLS, NMF-QMV, GMM, STRCM, and S^{3}TRM. Table III lists the quantitative results of endmembers and abundances for Urban.

As shown in Table III above, the proposed approach can obtain high-precision results whether in endmembers or abundances. It can be seen that compared to the geometric method—VCA and MVSA; statistical methods: NMF-sp, CNMF, and GMM; deep learning-based methods: DAEN, SNSA; and other topic-based methods: LDA, PLSA, and STRCM, the proposed methods can achieve higher accuracy in both SAD and RMSE. Although the proposed S^{3}TRM is slightly lower in endmember than DEpLSA, S^{3}TRM has a more significant improvement in abundances and time efficiency. Moreover, the results further illustrate the advantages of the proposed S^{3}TRM for complex images in unmixing. As shown in Fig. 12, the comparison of the abundance results by different methods shows that the proposed methods retain more information about the details of features. In comparison with the ground-truth abundance maps, S^{3}TRM can almost

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TABLE III

SAD ENDMEMBER ASSESSMENT AND RMSE ABUNDANCE ASSESSMENT FOR THE URBAN DATASET WITH FOUR ENDMEMBERS. NOTE THAT THE EXPERIMENTAL RESULTS SHOW BOTH NONTOPIC-BASED AND TOPIC-BASED METHODS, BEST RESULTS IN BOLD, RESPECTIVELY

		Non-topic-based methods							
		VCA-FCLS	MVSA	NMF-sp	CNMF	NMF-QMV	GMM	DAEN	SNSA
Urban	SAD	0.2941	0.3478	**0.2395**	0.2686	0.5429	0.4081	0.4304	0.4194
	RMSE	0.3134	0.3044	0.1284	0.2307	0.2963	0.3071	0.3048	**0.1126**
		Topic-based methods							
		LDA		PLSA		STRCM	DEpLSA		S^3TRM
	SAD	0.4091		0.3324		0.1710	**0.0803**		0.1230
	RMSE	0.2124		0.1748		0.1967	0.1197		**0.1123**

TABLE IV

COMPUTATIONAL TIME OF DIFFERENT METHODS

Datasets	**E**	**Time (seconds)**			
		NMF-sp	CNMF	DEpLSA	S^3TRM
Samson	3	58.02	26.79	38.46	**24.34**
Jasper	4	38.14	19.81	52.56	**7.67**
Urban	4	**73.32**	191.33	197.10	129.73

achieve consistent results for each feature, while other methods such as VCA-FCLS, NMF-QMV, and GMM have lower similarity in abundance estimation. S^3TRM stands out for both endmember extraction and abundance estimation, which demonstrates the effectiveness of the proposed methods.

F. Time Efficient Analysis With HSI

Comparison of computational time: to test the time complexity of S^3TRM, the traditional statistical methods were used as a comparison to test the running time on three real datasets. Table IV summarizes the calculation time required for NMF-sp, CNMF, DEpLSA, and S^3TRM models on the HSI.

Compared with NMF-based methods, it can be seen that S^3TRM shows more obvious advantages. For simple Samson dataset and Jasper dataset, S^3TRM has a faster unmixing rate than NMF-based methods. For complex Urban image, the running time of different algorithms has been greatly increased. Although the time required by S^3TRM is slightly higher than NMF-sp, the proposed method can obviously obtain higher unmixing accuracy. In addition, the model shows low model complexity and time consumption compared to the representative method DEpLSA which is based on the PTM. This further reflects that the proposed method can achieve a good balance between time and accuracy.

Overall, the proposed methods are experimentally verified to perform well in both qualitative assessment and quantitative evaluation. Although the geometrical-based method shows good performance in the extracting of endmembers for a simple dataset, due to mixing phenomena, spatial resolution, and other factors, pure spectral components are almost nonexistent in real hyperspectral scenes. And their estimation of abundances is certainly not as effective as endmembers. On the contrary, the proposed approaches are proved to exhibit better unmixing capabilities for real complex images in both SAD and RMSE. From the analysis of data complexity, for simple Samson data, the accuracy of S^3TRM has been significantly improved through the adaptive combination strategy of the STRCM with spectral information spatial information. For a more complex Jasper image, the accuracy of S^3TRM has achieved the best unmixing results. For the large and complex Urban image, the proposed S^3TRM has obvious advantages in both SAD and RMSE, indicating the applicability of the proposed strategy to complex scenes. The accuracy of S^3TRM is significantly better than geometric methods, deep learning-based methods, and other statistical methods. From the comparison of different unmixing methods, the STRCM is better than traditional PLSA and LDA, and the improved S^3TRM using sparse relaxation and group clustering is better than STRCM. On the whole, the S^3TRM is able to retrieve more stable endmember and abundance estimates both simple datasets and complex datasets, which further reflects the robustness of the proposed S^3TRM in actual application. The advantages are remarkable, especially for problems with complex images relatively. In addition, for all images in the experiment, the abundance maps obtained by S^3TRM and the real abundance maps have obtained relatively consistent results, which also confirms the rationality and effectiveness of the proposed methods.

V. CONCLUSION

In this research, an efficient unmixing framework called S^3TRM based on the topic model is proposed for HSIs unmixing. In S^3TRM, the topic model models the image by the statistics of the spectral reflectance of each pixel to obtain more semantic information. In this case, the sparse inference ability and features of fast convergence of sparse topic model are utilized to mine hidden endmembers and estimate fractional abundance in HSIs. In order to extract the endmembers in a more efficient way, we relax the topics by the sparsity of topic model to obtain a possible spectral representation of each endmember, thereby alleviate a series of problems caused by the endmember variability in HU. Then FCM is adopted to further extract endmembers and estimate abundances. By supplementing the spatial distribution information of the features, the accuracy of S^3TRM has been further improved. This study shows that the proposed S^3TRM has a better performance than the traditional topic model in finding high-quality semantic representation from HSIs, and has a lower model complexity. It achieved high SAD and

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RMSE scores in all real datasets, which demonstrates the unmixing advantages of S^{3}TRM in more complex scenes. As a result of the integration of the spatial information, the sparse prior and relaxation-clustering strategy, the proposed methods can obtain good unmixing results in both qualitative and quantitative evaluation. This further shows that unmixing by mining the semantic information of the HSI is effective. In future work, we plan to further exploit the framework by considering the automatic determination of the number of endmembers to improve the practicality of the proposed method.

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